

Scientific Computation

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Lecture 19 - The Singular Value Decomposition

Clarification: Rayleigh-Ritz

Last lecture we introduced the Rayleigh-Ritz quotient and proved two properties:

$$R_A(x) \triangleq \frac{x^T A x}{||x||^2}$$

- If (x, λ) are an eigenpair of A , then $R_A(x) = \lambda$.
- If x is an eigenvector of A , then x is an extremal point (maximum or minimum)

of $R_A(x)$.

- *Errata Corrige*: This holds only if A is a real **symmetric** matrix!

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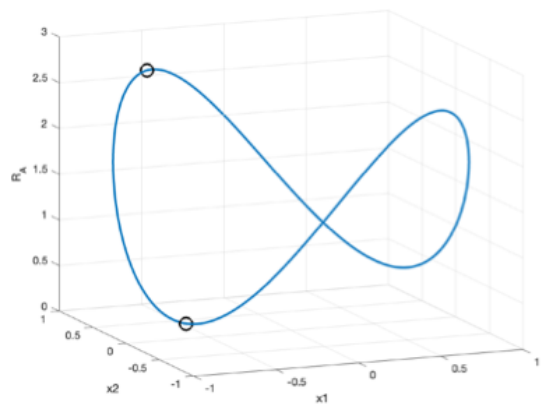
Clarification: Rayleigh-Ritz

Let's consider an example:

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

which has the following eigenpairs:

- First eigenpair: $\left(2.62, \begin{bmatrix} -0.53 \\ 0.85 \end{bmatrix} \right)$
- Second eigenpair: $\left(0.38, \begin{bmatrix} -0.85 \\ -0.53 \end{bmatrix} \right)$



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Lecture Outline

LECTURE OUTLINE

- Bauer-Fike bound
- The Singular Value Decomposition (SVD)
- Visualizing the SVD
- Applications of the SVD

Detecting Eigenpairs

When given an estimate of an eigenpair $(\hat{\lambda}_i, \hat{x}_i)$, we can check that they are at least close to being a true eigenpair via the idea of a residual:

$$r \triangleq A\hat{x}_i - \hat{\lambda}_i\hat{x}_i \approx 0.$$

The Bauer-Fike Theorem

Theorem 1: Let $A \in \mathbb{R}^{n \times n}$ be a diagonalizable matrix (i.e., $A = P\Lambda P^{-1}$ for some diagonal matrix of eigenvalues $\Lambda \in \mathbb{R}^{n \times n}$ and an invertible matrix $P \in \mathbb{R}^{n \times n}$). Let $(\hat{\lambda}, \hat{x})$ be an approximate eigenpair with residual

$$r = A\hat{x} - \hat{\lambda}\hat{x}.$$

Then there exists an eigenvalue $\lambda \in \sigma(A)$ such that

$$|\hat{\lambda} - \lambda| \leq \text{cond}_p(P) \frac{\|r\|_p}{\|\hat{x}\|_p}.$$

Proof of the Bauer-Fike Theorem

Assume that $\hat{\lambda}$ is **not** an actual eigenvalue of A (otherwise the proof is trivial). This means that $(A - \hat{\lambda}I)$ is nonsingular, and its inverse therefore exists. That means we can manipulate the expression for the residual as follows:

Proof of the Bauer-Fike Theorem Continued

Note that $(\Lambda - \hat{\lambda}I)^{-1}$ is a diagonal matrix. Therefore, its p -norm is easy to compute:

A More Useful Corollary

Corollary 1: If A is symmetric, the bound becomes

$$|\hat{\lambda} - \lambda| \leq \frac{\|r\|_p}{\|\hat{x}\|_p}.$$

Example Eigenpair

Example 1: Consider the symmetric matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

and consider the approximate eigenpair $(1.01, [0.5 \ 0.5]^\top)$.

Visualizing the Bound

Example 1 cont.: What happens if we systematically perturb a true eigenpair?

```
using Plots

n_plot = 100
θ = range(0.0, stop=π/2, length=n_plot)
λ = 1.0
dλ = range(0.0, stop = 1.0, length=n_plot)

bauer_fike(θ, dλ) = norm((A - (λ+dλ)*I)*[cos(θ); sin(θ)])
bound = bauer_fike.(θ', dλ)

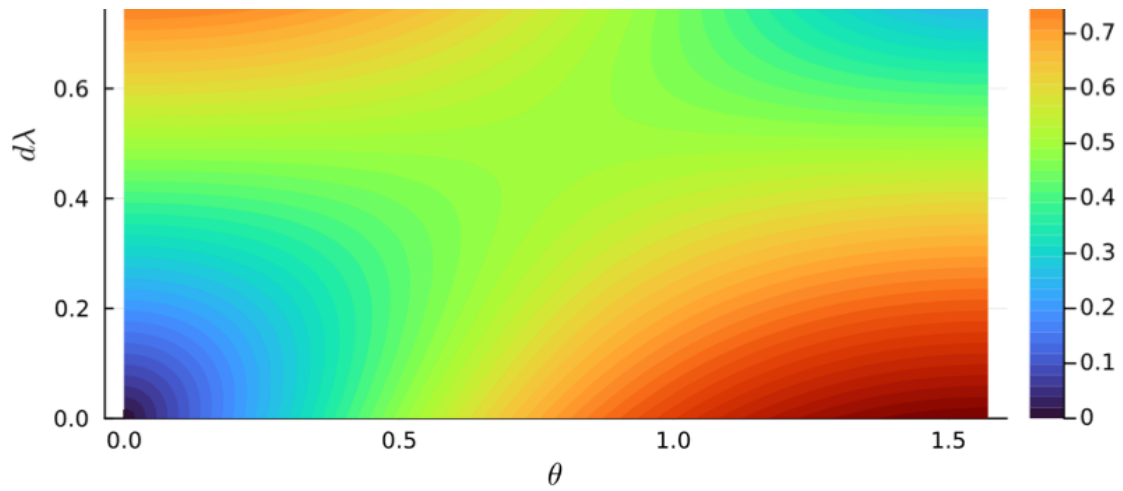
contourf(θ, dλ, bound, fill=true, color=:turbo,
         lw=0.0, aspect_ratio=:equal, levels=50)
axis!("\$\\theta\$")
axis!("\$d\\lambda\$")
title!("Bauer-Fike Bound")
```

Visualizing the Bound

Example 1 cont.: The resulting contour plot is interesting:

Bauer-Fike Bound





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Lecture Outline

- Bauer-Fike bound
- **The Singular Value Decomposition (SVD)**
- Visualizing the SVD
- Applications of the SVD

$$A = U \Sigma V^T$$

Rectangular Matrices

Rectangular Matrices

Eigenvalues and eigenvectors are great, but they only exist for *square* matrices! Is there a similar concept for $A \in \mathbb{R}^{m \times n}$ when $m \neq n$?

$$M \triangleq A^T A \in \mathbb{S}_+^n \quad \Leftrightarrow \quad \begin{matrix} M \geq 0 \\ M^T = M \end{matrix}$$

$$\hookrightarrow \text{positive semidefinite!} \quad x^T M x \geq 0 \quad \forall x \\ = (x^T A^T) A x$$

$$y \triangleq A x \Rightarrow x^T M x = y^T y \\ = \sum_{i=1}^n y_i^2 \geq 0$$

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Spectral Theorem for $A^T A$

If $A \in \mathbb{R}^{m \times n}$, what can we say about the matrix

$$M = A^T A \in \mathbb{S}_+^n$$

All eigenvalues of M are nonnegative ($\lambda_i \geq 0$, $\forall \lambda_i$ in spectrum of M)

$$M = Q \Lambda Q^T, \quad Q^T Q = I \quad (\text{orthogonal})$$

$$\sigma_i = \sqrt{\lambda_i} \geq 0$$

$\uparrow$ singular value of A

$$\Lambda = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

$\hookrightarrow$ eigenvalues of M

Spectral Theorem for AA^T

What about the matrix

$$M' = AA^T \in \mathbb{R}^{m \times m}?$$

$\Rightarrow$ same properties as previous slide, except an $m \times m$ symmetric PSD matrix: $M' = Q \Lambda' (Q')^{-1}$

The Singular Value Decomposition

Theorem: We can write any matrix $A = \mathbb{R}^{m \times n}$ as

$$A = U \Sigma V^T,$$

$$\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \end{bmatrix}$$
 Rectangular diagonal matrix

where $U \in \mathbb{R}^{m \times m}$ is orthogonal, $\Sigma \in \mathbb{R}^{m \times n}$ is a diagonal (possibly rectangular)

matrix, and $V \in \mathbb{R}^{n \times n}$ is orthogonal.

$$A = \overset{n}{U} \underbrace{\overset{n}{\begin{bmatrix} \sigma_1 & \sigma_2 & 0 & \\ 0 & \ddots & \sigma_m & 0 \end{bmatrix}}_{\bar{\Sigma}} \overset{n}{V^T}$$

$\sigma_i \geq 0$

A has 3 factors :
 1. rotation / reflection (V^T)
 2. scaling ($\bar{\Sigma}$)
 3. rotation / reflection (U)

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A Simple Example

Example: What is the SVD of the matrix

$$\sigma_1 = 1 = \sigma_2 > 0 \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}?$$

$$A = U \bar{\Sigma} V^T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↘ V^T is a reflection across the x - z plane.

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SVD with Julia

Example 1: Julia can easily compute the *thin* SVD:

```
using LinearAlgebra
A = [1 0 0;
     0 -1 0]
U, Σ, Vt = svd(A)
display(U)
display(diagm(Σ))
display(Vt')
```

$$k = \min\{m, n\}$$

$$\Rightarrow \Sigma \in \mathbb{R}^{k \times k}$$

$$V \in \mathbb{R}^{k \times n}$$

$$(\text{if } k = m):$$

$$U \in \mathbb{R}^{m \times k}$$

2x2 Matrix{Float64}:

```
1.0  0.0
0.0  1.0
```

2x2 Matrix{Float64}:

```
1.0  0.0
0.0  1.0
```

2x3 Matrix{Float64}:

```
1.0  0.0  0.0
0.0 -1.0  0.0
```

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Relating Singular Values of A to Eigenvalues of $A^T A$

Let's check the bold claim we made earlier about the singular values σ_i of A being the square root of the eigenvalues of $A^T A$:

$$A = U \Sigma V^T$$

$$\begin{aligned} \Rightarrow M = A^T A &= (U \Sigma V^T)^T U \Sigma V^T \\ &= V \Sigma^T (U^T U) \Sigma V^T \\ &= V (\Sigma^T \Sigma) V^T \\ &= V \Lambda V^T = \text{eigen decomposition of } M \end{aligned}$$

$V^T = \text{orthogonal}$

Repeat for $M' = A A^T$, we see that "singular vectors" in column of U are just eigenvector of M' !

$$\sigma_i = \sqrt{\lambda_i} \text{ for } \lambda_i \text{ in spectrum of } M$$

Lecture Outline

- Bauer-Fike bound
- The Singular Value Decomposition (SVD)
- **Visualizing the SVD**
- Applications of the SVD

Visualizing the SVD

Example 2: Let's use Julia to visualize a more interesting example. Consider the matrix

$$A = \begin{bmatrix} 0.5 & 0.2 \\ 1 & 1 \end{bmatrix}.$$

The SVD of A is

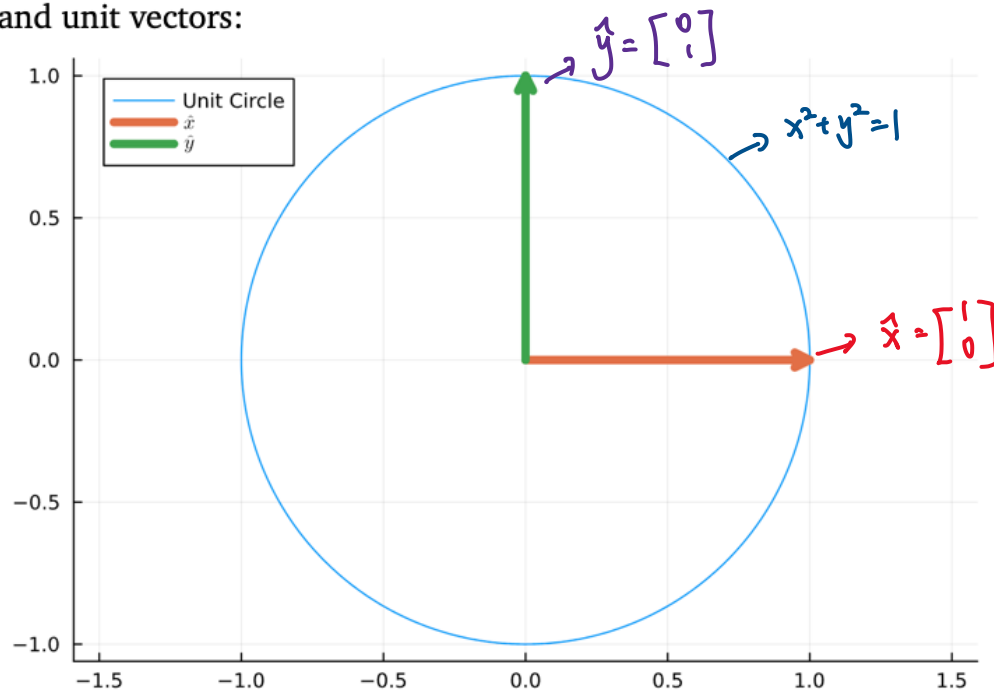
$$A = U\Sigma V^T \approx \begin{bmatrix} -0.34 & -0.94 \\ -0.94 & 0.34 \end{bmatrix} \begin{bmatrix} 1.5 & 0 \\ 0 & 0.2 \end{bmatrix} \begin{bmatrix} -0.74 & -0.67 \\ -0.67 & 0.74 \end{bmatrix}.$$

$\nwarrow$ 3. rotation / reflection (U) $\searrow$ 1. rotation / reflection (V^T)
 $\swarrow$ 2. scaling along elementary basis vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (Σ)

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Step 1: Plot a Unit Circle and Orthogonal Unit Vectors $\hat{x}$ and $\hat{y}$

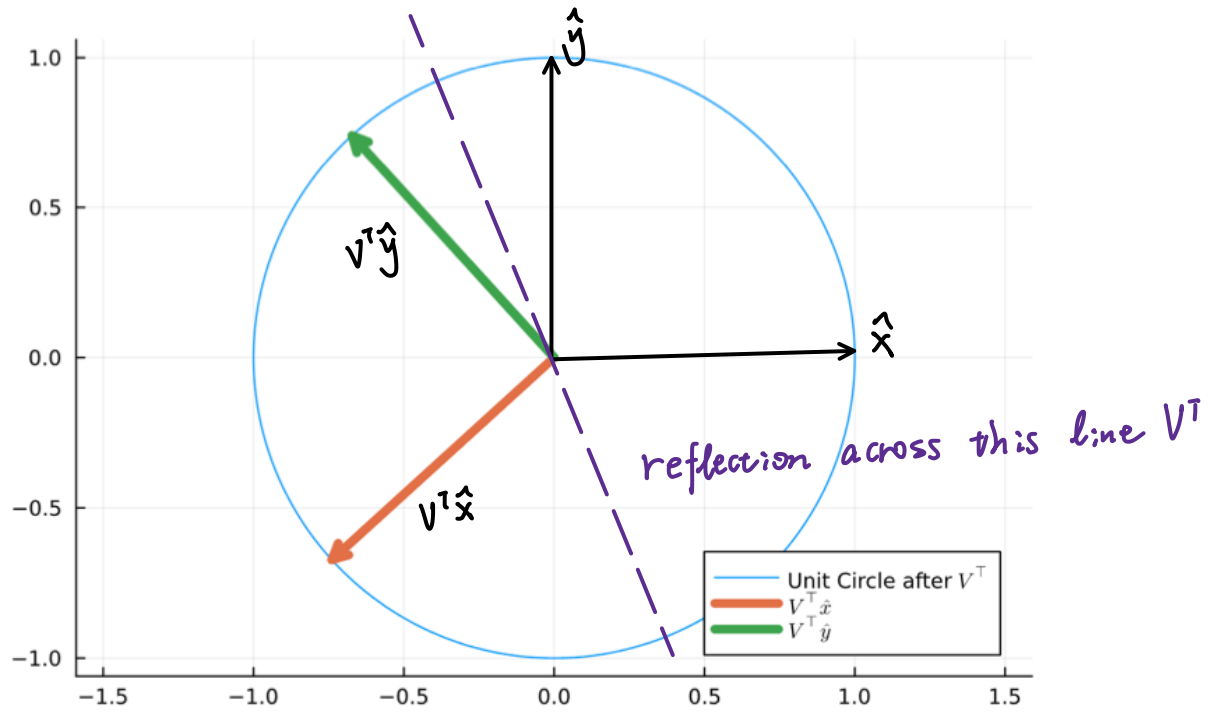
Example 2 cont: We will demonstrate what the SVD reveals about the effect A has on vectors by plotting the effect of each component (V^T , Σ , and U) on the unit circle and unit vectors:



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Step 2: Apply V^T

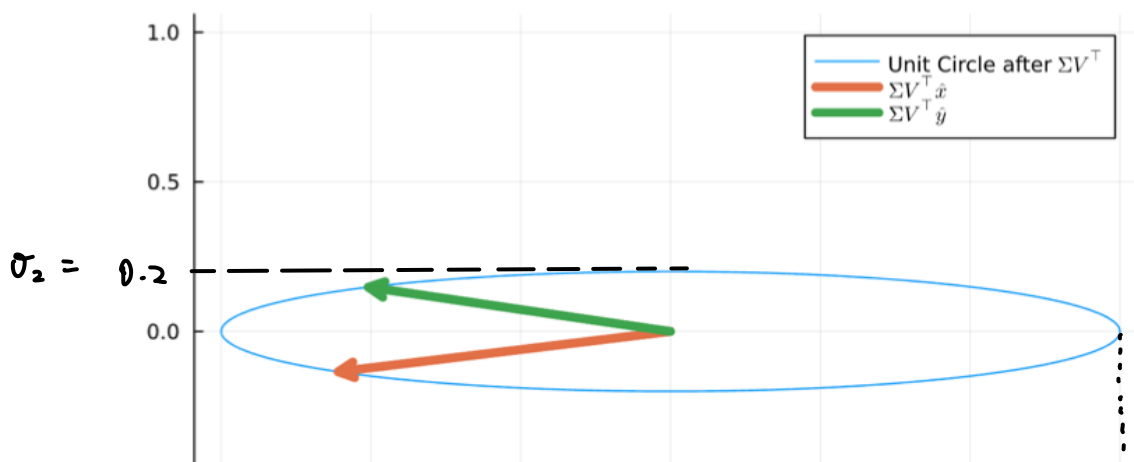
Example 2 cont: We begin (from right to left) with $V^T x$, which is a **reflection**:

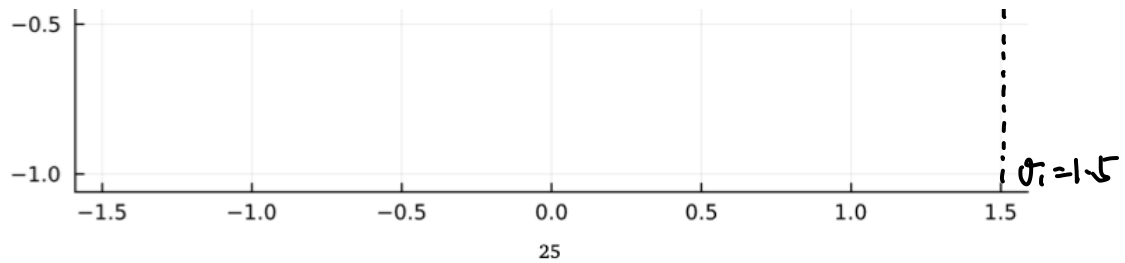


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Step 3: Apply Σ

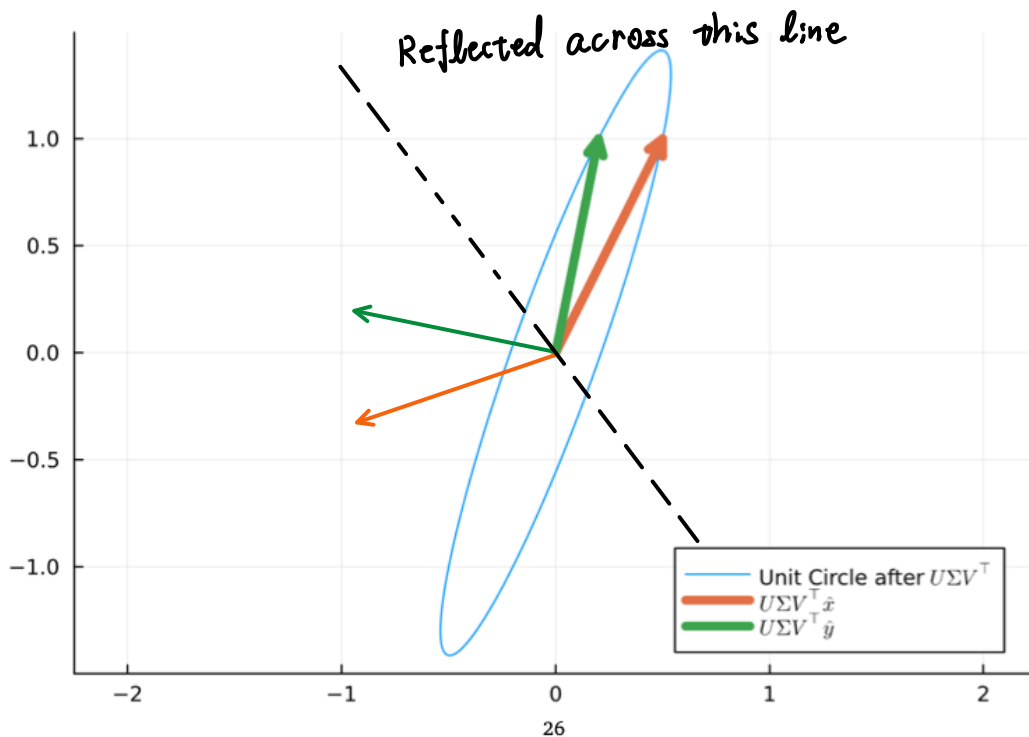
Example 2 cont: Next we have $\Sigma V^T x$, which **scales** the elements of $V^T x$:





Step 4: Apply U

Example 2 cont: Finally we have $U\Sigma V^T x = Ax$, which reflects $\Sigma V^T x$:



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Computing the Moore-Penrose Pseudoinverse

Recall the Moore-Penrose pseudoinverse of $A \in \mathbb{R}^{m \times n}$, which is a matrix $A^\dagger$ satisfying:

$$AA^\dagger A = A$$

$$A^\dagger AA^\dagger = A^\dagger$$

$$(AA^\dagger)^\top = AA^\dagger$$

$$(A^\dagger A)^\top = A^\dagger A$$

} symmetric

rank = # of linear independent
col/rows of A
= # non zero singular values

we saw that when $m > n$ (tall matrix)
and $\text{rank}(A) = n$ (full rank), we have an overdetermined system and
the pseudo inverse is: $A^\dagger (\underbrace{A^\top A}_{SS^\top})^{-1} A^\top$

Pseudoinverse for Full Row-Rank

Conversely, if $m < n$ but A is still full rank, we have

$$A^\dagger = A^T(AA^T)^{-1},$$

which is derived as part of the solution to Problem 2 in Practice Problem Set 3.

This helps us solve underdetermined $Ax=b$ for the smallest (in terms of $\|x\|_2$) x .

Can we use the SVD to generalize the pseudo inverse beyond these 2 cases? I.e. when A is not full rank.

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Pseudoinverse of a Diagonal Matrix

Let $D \in \mathbb{R}^{m \times n}$ be a diagonal matrix (i.e., $D_{ij} = 0 \forall i \neq j$). What is $D^\dagger$?

$$D = \begin{bmatrix} d_1 & & 0 & \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n & \\ & & & & 0 \end{bmatrix} \text{ if } m < n$$

$$D = \begin{bmatrix} d_1 & & 0 \\ & d_2 & \\ & & \ddots \\ 0 & & & d_n \\ & & & & 0 \end{bmatrix} \text{ if } m > n$$

$$\text{Try: } (D^\dagger)_{ii} = \begin{cases} \frac{1}{D_{ii}}, & \text{if } D_{ii} \neq 0 \\ 0 & \text{else} \end{cases}$$

$$\text{and } (D^\dagger)_{ij} = 0$$

Putting it All Together

We can now show that the pseudoinverse of $A = U\Sigma V^T$ is simply

$$A^\dagger = V\Sigma^\dagger U^T.$$

E.g. show property $A^\dagger A A^\dagger = A^\dagger$ from earlier:

$$\begin{aligned} A^\dagger A A^\dagger &= \underbrace{V \Sigma^\dagger \Sigma V^T}_{A^\dagger A} (V \Sigma^\dagger U^T) \\ A^\dagger A &= V \Sigma^\dagger U^T (U \Sigma V^T) \\ &= V \Sigma^\dagger \Sigma V^T \\ &= V (\Sigma^\dagger \Sigma \Sigma^\dagger) V^T = V \Sigma^\dagger V^T = A^\dagger \end{aligned}$$

Point Cloud Alignment

Consider the problem of aligning two noisy measurements of **points clouds** with known correspondences between them. This is a fundamental problem in computer vision and robotics.



Figure from "Efficient global point cloud alignment using Bayesian nonparametric mixtures." Straub et al., IEEE CVPR 2017.

这一章讲的是奇异值分解。

16页讲了一个问题，即只有square matrix才有特征值，但是当A不是square matrix，那我们如何定义特征值？

我们可以构造一个square matrix，比如 $M=A^T A$ ，同时它是 semidefinite的。

17页指出，

3. The Singular Value Decomposition (SVD)

大定理：

对任意 $A \in \mathbb{R}^{m \times n}$ ，存在分解

$$A = U \Sigma V^T$$

其中：

- $U \in \mathbb{R}^{m \times m}$ 是正交矩阵（列向量叫做左奇异向量）
- $V \in \mathbb{R}^{n \times n}$ 是正交矩阵（列向量叫做右奇异向量）
- $\Sigma \in \mathbb{R}^{m \times n}$ 是“对角但可能是矩形”的矩阵：

$$\Sigma = \begin{bmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & 0 \\ & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

其中 $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, $r = \text{rank}(A)$

其中 $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, $r = \text{rank}(A)$

几何解释（非常重要）：

- V^T : 把输入空间旋转/翻转到“好对齐的坐标轴”
- Σ : 沿这些坐标轴独立地拉伸或压扁（按 σ_i 缩放），并可能丢掉某些方向（对应 0 奇异值）
- U : 把结果再旋转/翻转到输出空间

你可以记成：

“旋转 → 缩放 → 再旋转”

情况	矩阵形状	零块位置	非零奇异值数量
$m > n$	tall (高矩阵)	下方是零行	$r = n$
$m < n$	wide (宽矩阵)	右侧是零列	$r = m$

4. Simple Example（课上例题）

矩阵：

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \quad (2 \times 3)$$

结论他们给的是：

$$A = U\Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

解释：

- $\sigma_1 = \sigma_2 = 1$, 奇异值都是 1
- $V = I_3$: 输入坐标系不需要旋转
- Σ : 从 3D 只保留前两个方向（相当于丢掉 z 分量）
- U : 把第二个轴翻个号（反射）

几何上, A 做的事情是：

$(x, y, z) \mapsto (x, -y)$ 。

也就是“投影到 x-y 平面，再把 y 翻转”。

20页指出, $M = V\Lambda V^T$, 根据spectral theorem, 我们知道V的column vector是M的eigenvector; 同时, 对于 $M=AA^T$, 此时U的column vector是M的eigenvector。

这一页通常会说：

- 如果 $\sigma_i > 0$, 对应的是有效方向（这些方向的输入会被送到输出）
- 如果 $\sigma_i = 0$, 那些方向会被“杀死”成 0, 说明这些方向属于 null space

也就是说：

- nonzero 奇异值的个数 = $\text{rank}(A)$
- A 的列空间 (image / range) 由对应的左奇异向量 u_i 生成
- A 的零空间 (null space) 由对应于 $\sigma_i = 0$ 的右奇异向量 v_i 生成

这部分很重要，因为它告诉你：

SVD 不只是分解，它还直接告诉你矩阵的秩、列空间、零空间。

21页开始是可视化展示SVD：

7. Visualizing the SVD (可视化步骤)

这几页是 2D/3D 画图，展示几何意义。流程基本是：

Step 1: start with unit circle / unit sphere

- 画一个单位圆/球，表示“所有长度为1的向量”

Step 2: Apply V^T

- V^T 只是旋转/反射，不改变长度，所以单位圆还是圆，只是坐标轴转了（基底换了）

Step 3: Apply Σ

- Σ 会按 $\sigma_1, \sigma_2, \dots$ 不同地拉伸每个主方向
- 原来的圆变成椭圆（在 3D 就是椭球/扁的片）
- 每个半轴长度 = 对应的奇异值 σ_i

这是核心直觉：**奇异值 = 主拉伸因子**

Step 4: Apply U

- 最后 U 再把这个扁平的椭圆旋转到输出空间的方向

所以整套 SVD 变换是：

“先把坐标轴对正，再按主轴方向压扁拉长，再转回去”。

其实SVD就是之前eigendecomposition $A = P\Lambda P^{-1}$ 的推广，只是特征值分解只能用在square matrix，而SVD可以用在任何矩阵。

它们的目的都是，想要知道矩阵 A 在每个方向上拉伸程度（这就需要分析eigenvector和eigenvalue）。但是，由于 A 不一定是square matrix，所以特征值和特征向量在这里不一定适用了，不一定存在一个特征向量只被拉伸方向不变。因此，我们换一种方法：

假设 A 为 $m \times n$ ， Σ 是 $m \times n$

V 是输入空间的一组 $n \times n$ 的正交基组成的正交矩阵， $AV = \Sigma U$ ， U 是输出空间，是 $m \times m$ 的正交矩阵，由 m 个输出空间的正交基组成。 Σ 是拉伸系数。输入空间由 V 的列向量span，输出空间由 U 的列向量span。**所以我们现在结合了寻找 A 变换后仍然正交的正交向量组和寻找拉伸方向的意义！**

注意，由于V是orthogonal的，所以 $V^{-1}=V^T$ 。注意 Σ 中的singular value σ 是对于输入维度n来说的。

n维输入空间的正交基集合V，经历了A变换，映射到m维空间的正交基集合U，但是只是被拉伸了 σ 倍， σ 叫做singular value。所以SVD就是告诉你，输入空间的一组正交基，怎么被拉伸，拉伸完之后指向哪。

SVD的一个重要应用就是算pseudoinverse:

$$\begin{aligned} A &= U\Sigma V^T \\ A^+ &= V\Sigma^+ U^T \\ \Sigma_{ij}^+ &= \begin{cases} 1/\sigma_i, & \text{if } i=j \leq r \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

例题:

Ex. $A = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$, find SVD.

$$A = U\Sigma V^T$$

$$A^T A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} = V(\Sigma^T \Sigma)V^T$$

$$\lambda_1 = 4, \lambda_2 = 2$$

$$\therefore \sigma_1 = 2, \sigma_2 = \sqrt{2}$$

$$\therefore \text{Solve for } A^T A x = \lambda x,$$

$$\begin{bmatrix} 2-4 & 0 \\ 0 & 4-4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (\lambda_1)$$

$$\begin{bmatrix} 2-2 & 0 \\ 0 & 4-2 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (\lambda_2)$$

$$\therefore V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \therefore \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

$$A A^T = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Solve } (A A^T - \lambda I) = 0$$

$$\therefore \begin{vmatrix} 4-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0 \Rightarrow (4-\lambda)(1-\lambda) = 0$$

$$\begin{vmatrix} 0 & 1-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{vmatrix} = 0 \quad \therefore (4-\lambda)(1-\lambda)^{-1} \\ (4-\lambda)\lambda(\lambda-2) = 0 \\ \lambda_1 = 4, \lambda_2 = 2, \lambda_3 = 0$$

$$\therefore \text{For } \lambda_1 = 4 \Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 1 & -3 \end{bmatrix} u_1 = 0, u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{For } \lambda_2 = 2 \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} u_2 = 0, u_2 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\lambda_3 = 0 \Rightarrow u_3 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Note: The size of Σ is same as A , for the calculation of σ , just calculate the λ for smaller matrix in $\{V^T V, VV^T\}$ for here is $A^T A$.

The first column of Σ rep. σ_1 , and the U and V 's first column also need to correspond to λ_1 's eigenvector.